605.486 – Android Development

Module 18
Preferences

Shared Preferences

Basic Shared Preference Usage

```
// for activity or preference subsets
SharedPreferences preferences = getPreferences(MODE_PRIVATE);
// same as
// getSharedPreferences(getClass().getSimpleName(), MODE_PRIVATE);

// or for application-level preferences
// NOTE: THIS IS WHERE PreferenceFragments STORE PREFS!
SharedPreferences preferences2 =
        PreferenceManager.getDefaultSharedPreferences(this);

String name = preferences.getString("name", "nameless");

Editor editor = preferences.edit();
editor.putString("name", "Scott");
editor.commit();
```

Preference Screen

- Xml file defines <PreferenceScreen>
- Defines widgets to collect data
- Often references arrays of data

```
<string-array name="statesLong">
    <item>Alabama</item>
</string-array>
```

Preference Screen

- PreferenceActivity
- PreferenceFragment (preferred after Honeycomb)

```
SharedPreferences preferences =
    PreferenceManager.getDefaultSharedPreferences(
                                                  context);
preferences.getString(name, defaultValue);
preferences.getBoolean(name, defaultValue);
…
```

- Note: Does not show current values!

Example 1 – Basic Pref Screen

- prefs.xml
- MyPrefsFragment
- MyPrefsActivity

- Demonstrate multi-fragment prefs

Example 2 - Preference Screen Update

- Add SharedPreferenceChangeListener
- Updates summary text
- Register in onResume
- Unregister in onPause

Example 3 – Uninstall, Poof!

- Preferences are deleted when the application is uninstalled!
- What if user wants to reinstall and keep their settings?

Example 4 – Preference Backup

- Define a Backup Agent
  - Stores shared preferences or files in cloud
  - BackupManager.dataChanged()
  - Automatic reinstall
  - Requires “Back up my data”
  - Not on all devices!
  - NOT IMMEDIATE!
- Need to register for key
- bmgr util for dev – force immediate run